

ABHINAV JAVA

Research Fellow, MICROSOFT RESEARCH

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RESEARCH EXPERIENCE

MICROSOFT RESEARCH, India Research Fellow with Amit Sharma , Nagarajan Natarajan , Srivathsan Koundinyan , Vineeth N Balasubramanian	08/2024 – Present
UNIVERSITY OF VIRGINIA, Visiting Researcher with Chirag Agarwal	03/2024 – 03/2025
ADOBE RESEARCH, India Research Associate I, II with Balaji Krishnamurthy Research Intern with Balaji Krishnamurthy	05/2021 – 08/2024
MASSACHUSETTS INSTITUTE OF TECHNOLOGY, Collaborator with Ayush Chopra , Ramesh Raskar	03/2021 – 10/2022

EDUCATION

DELHI TECHNOLOGICAL UNIVERSITY, India B.Tech. in Computer Engineering > Research Excellence Award 2022.	08/2018 – 07/2022
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PUBLICATIONS

* DENOTES JOINT FIRST AUTHORSHIP

UNDERSTANDING TASK TRANSFER IN VLMS

Bhuvan Sachdeva*, Karan Uppal*, [Abhinav Java](#)*, Vineeth N. Balasubramanian
CVPR '26 ([ORAL](#)), UniReps@NeurIPS '25

CHARACTERIZING DEEP RESEARCH: A BENCHMARK AND FORMAL DEFINITION

[Abhinav Java](#)*, Ashmit Khandelwal*, Sukruta Midigeshi*, Aaron Halfaker, Amit Deshpande, Navin Goyal, Ankur Gupta,
Nagarajan Natarajan, Amit Sharma
ICLR '26, SEA@NeurIPS '25

FRUGALRAG: LESS IS MORE IN RL FINETUNING FOR MULTI-HOP QA

[Abhinav Java](#), [Srivathsan Koundinyan](#), [Nagarajan Natarajan](#), [Amit Sharma](#)
ICLR '26, ER@NeurIPS '25

TOWARDS OPERATIONALIZING RIGHT TO DATA PROTECTION

[Abhinav Java](#)*, [Simra Shahid](#)*, [Chirag Agarwal](#)
NAACL (Main) '25, MLDPR@ICLR '25

REEDIT: MULTIMODAL EXEMPLAR-BASED IMAGE EDITING

Ashutosh Srivastava*, Tarun Ram*, [Abhinav Java](#)*, Avadhoot Jadav, Silky Singh, Surgan Jandial, Balaji Krishnamurthy
WACV '25, AI4VA@ECCV '24

ALL SHOULD BE EQUAL IN THE EYES OF LMS: COUNTERFACTUALLY AWARE FAIR TEXT GENERATION

[Pragyan Banerjee](#)*, [Abhinav Java](#)*, [Surgan Jandial](#)*, [Simra Shahid](#)*, [Shaz Furniturewala](#), [Balaji Krishnamurthy](#), [Sumit Bhatia](#)
AAAI '24

THINKING FAIR AND SLOW: ON THE EFFICACY OF STRUCTURED PROMPTS FOR DEBIASING LANGUAGE MODELS

[Shaz Furniturewala](#), [Surgan Jandial](#), [Abhinav Java](#), [Pragyan Banerjee](#), [Simra Shahid](#), [Sumit Bhatia](#), [Kokil Jaidka](#)
EMNLP (Main) '24

ONE-SHOT DOC SNIPPET DETECTION: POWERING SEARCH IN DOCUMENTS BEYOND TEXT

[Abhinav Java](#), [Shripad Deshmukh](#), [Milan Agarwal](#), [Surgan Jandial](#), [Mausoom Sarkar](#), [Balaji Krishnamurthy](#)
WACV '23

LEARNING TO CENSOR BY NOISY SAMPLING

[Ayush Chopra](#), [Abhinav Java](#), [Abhishek Singh](#), [Vivek Sharma](#), [Ramesh Raskar](#)
ECCV '22

PATENTS

Relational Loss for Text based Style Transfer, US Patent 20250349050A1 .	2025/11/13
Personalized form error correction propagation, US Patent 12394236 .	2025/08/19
One Shot Doc Snippet Detection, US Patent 20250005048A1 .	2025/01/02
Form structure similarity detection, US Patent 12124497B1 .	2024/10/22
Systems and methods for data augmentation, US Patent 20240119122A1 .	2024/04/11

FEATURED PROJECTS AND COLLABORATIONS

EFFICIENT MULTI-HOP REASONING FOR RETRIEVAL AUGMENTED GENERATION w/ Amit Sharma , Nagarajan Natarajan , Srivathsan Koundinyan	08/2024 – Present Microsoft Research
> Led the development of a project to create efficiently adaptable and low-latency RAG system with SLMs. > Proposed FrugalRAG: by learning efficient use of test-time compute using reinforcement learning > Improved latency and data efficiency by approximately 2x.	
RETHINKING TASKONOMY IN THE ERA OF LARGE VISION LANGUAGE MODELS w/ Vineeth N Balasubramanian	04/2025 - Present Microsoft Research
> Proposed a framework to study transferability of perception tasks in Vision-Language Models (VLMs). > Introduced a novel metric to quantify cross-task influence in VLM finetuning. > Revealed actionable insights – beneficial task cliques and transfer trends for effective finetuning strategies.	
A MODEL AGNOSTIC APPROACH TO CREATE UNLEARNABLE TEXT DATASETS w/ Chirag Agarwal	03/2024 – 03/2025 University of Virginia
> Developed the first model-agnostic, efficient unlearnable text data generation framework for user data privacy. > Proposed RegText: a mechanism to perturb sensitive datasets using information-theoretic insights. > Demonstrated that finetuning large models (e.g. GPT-4o) on our datasets drops performance below zero-shot!	
PROTECTING PRIVATE INFORMATION IN 3D POINT CLOUDS FOR PERCEPTION TASKS w/ Ayush Chopra , Ramesh Raskar	03/2021 – 10/2022 Camera Culture, MIT Media Lab
> Designed an approach to simplify the release of sensitive point cloud data with privacy protection. > Implemented CBNS: an end-to-end differentiable sampler and a novel loss to protect point clouds from adversaries. > Demonstrated improved privacy-utility tradeoffs with CBNS on perception tasks on 3D face datasets.	

HONORS AND AWARDS

CVPR ORAL, 2026	Research selected for Oral presentation at CVPR 2026.
ICLR TRAVEL GRANT, 2026	Received travel grant for 1600 USD to present my research at ICLR'26 in Rio De Janeiro.
NEURIPS SCHOLAR AWARD, 2025	Received travel grant to present my research at NeurIPS'25 in San Diego.
AAAI ORAL TALK, 2024	Selected for an invited talk at AAAI 2024, Student Abstract Track.
RESEARCH EXCELLENCE AWARD, 2022	Awarded for undergraduate research during undergrad (1 in approx 2000).
SEMESTER MERIT AWARD, 2021	For securing University Rank 1 in the 5th Semester (1 in approx 2000).
JOINT ENTRANCE EXAMS, 2018	Secured 99.28 percentile in Main and 96.78 percentile in Advanced (in 1.2 million).
NATIONAL TALENT SEARCH EXAM, 2017	Qualified Stage 1 (Top 1% in 1.2 million).

LEADERSHIP, INVITED TALKS, AND SERVICE

FIRESIDE CHAT, LiveDRBench . Plutos Dev	Fall 2025
CAREER TALK, AI Summer School . Delhi Technological University	Summer 2025
PRE-PLACEMENT TALK, Adobe Hiring . Delhi Technological University	Summer 2024
TEAM UAS-DTU, Vice Captain, Software & Airframes Technician.	2019 - 2020
> Secured a funding of \$1M from Adani Group for building Drone Swarm for disaster relief. (Media) > Awarded the best communication architecture prize of INR 250K in IAF National Competition. (Media)	
UMEED FOUNDATION, Volunteer.	2022 - 2023
> Provided career counseling to college students, and assisted with interview preparation. > Taught middle school students basic science and mathematics during covid. > Awarded for my volunteering efforts by Adobe as one of the top 50 contributors.	
Reviewer: CVPR '26, ICLR '26, CVPR '25, KDD '24, CVPR'24, U&ME@ECCV'24, WACV'23, ECCV'22	